Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs <- rhfs |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSPACE, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECCOMMREN, RNTAPPFEEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINCTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
   )

# Renaming some factor levels for later convenience
rhfs <- rhfs |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
```

```
      "Assisted living limited interview; self response" = "3",
      "Conversion from partial to full interview" = "5",
      "Interview Compl; field interview" = "201",
      "Assisted living limited interview; field interview" = "203",
      "Sufficient partial; field interview" = "204",
      "Military limited field interview" = "206"
    )
  )

# Rearranging variables greatly improves View(rhfs)
questionnaire_order <- c(
  # 1c-1e and property definition
  "PROPTOWN", "PROPCON", "COMPLEX",
  "NUMBLD_R", "NUMUNITS_R", "NUMCAT_R",

  # 2-4e
  "UNITSTAT", "YRNEWBLG_R", "HOWACQ", "YRPROPACQ_R", "PURPRICE_R",
  "CASHDWN", "CASHDWNAMT", "MRKTVAL_R", "HOWMVAL", "APPRWAIVER",

  # 5-6: occupancy, rents, and rent payment methods
  "BROOMS", "TOTRENTVAC", "TRENOC_R", "TRENVA_R", "TOWNOC_R",
  "TEMPOC_R", "TOTHVA_R", "LOWRENT", "HIRENT", "RENT",
  "RENTNO_R", "RENTLT249_R", "RENT274_R", "RENT299_R", "RENT349_R",
  "RENT399_R", "RENT449_R", "RENT499_R", "RENT549_R", "RENT599_R",
  "RENT649_R", "RENT699_R", "RENT749_R", "RENT799_R", "RENT899_R",
  "RENT999_R", "RENT1099_R", "RENT1249_R", "RENT1499_R", "RENT1999_R",
  "RENT2499_R", "RENT2999_R", "RENT3499_R", "RENT3999_R", "RENT4499_R",
  "RENT4999_R", "RENT5000_R", "BED0_R", "BED1_R", "BED2_R", "BED3_R",
  "PAYONLINE", "PAYTHIRD", "PAYAPP", "PAYCREDIT", "PAYCHK", "PAYCASH", "PAYOTH",

  # 6a-11c
  "LEASLNG", "LNGTRMPRV", "WEEKLNG", "WEEKLNGNUM", "WEEKLNGPCT",
  "RNTAPPFEE_R", "NORNTAPPFEE", "RNTAPPFEEWAIV", "PCTFEEWAIV_R", "SECDEPAMT",
  "CONCSS", "CONCSSNUM", "SENIOR", "RCONTROL", "RCONTROLAHS",
  "OFFSTPARK", "OFFSTPARKAHS", "IELEC", "IGAS", "IWATER", "ISEWER", "ITRASH",
  "ICABLE", "IWIFI", "IPOOL", "IPARKING", "PROPANS",

  # 12a-15e
  "TOTCOLL", "COMMSPACE", "RECCOMMREN", "COMMRENT", "MNGMNT", "HRSMNGMNT",
  "ADVERTISE", "ADWEBSITE", "ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
  "ADAGENCY", "ADAGENT", "MLS",

  # 16: operating expenses
  "OPPAY", "OPTAX_R", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
  "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX", "OPEX_R",

  # 17-18c: capital expenses and ownership
  "CAPEX_R", "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT", "CPACCAMT",
  "CPOTHAMT", "CPRESAMT_R", "OWNENT", "OWNOTHPROP", "NUMOTHPROP", "NUMOTHUNITS",

  # 19-23: subsidies, tax credits, and tax reductions
  "HUDUNITS", "GOVBELOW", "RSUBSIDY", "ELDERLY", "GOVGRANT", "BENEOTH",
```

```
  "LIHTCUNITS", "STLOWINC", "HISTORIC", "LOWINCTEN", "LOWINAREA", "ECONDEVLP", "OTHRCV",

  # 24-37: debt/mortgage section and HUD voucher history
  "DEBT", "WHYNOMRT", "DEBTNUM", "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R",
  "APPVAL_R", "GREENMORT", "MORTGET", "LOCTYP", "REGPAY", "PMTAMT_R", "PMTFREQ",
  "RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX", "ACCEPT"
)

rhfs <- rhfs |>
  relocate(
    CONTROLPUF, OUTCOME, SAMPSTRATA, SURVYR,
    all_of(questionnaire_order)
  ) |>
  # Keep the full-sample and replicate weights together at the far right.
  relocate(
    WEIGHT, starts_with("REPWGT"), .after = last_col()
  )
```

```
# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )
```

```
## # A tibble: 1 x 3
##   properties properties_se   moe
##        <dbl>         <dbl> <dbl>
## 1     18965.          165.  271.
```

```
# Makes View(rhfs) much more productive.
rhfs <- rhfs |>
  select(-starts_with("REPWGT"), -WEIGHT,
         -SURVYR, -CONTROLPUF, # is always 2024, ID isn't useful in data analysis
         # J flag variables mark values that were manually reviewed; note that
```

```
        # they may become relevant if outliers are under scrutiny.
        -starts_with("J"),
        # A continuous variable coded as a factor without documentation; may
        # be useful for pure prediction
        # -PCTFEEWAIV_R,
        # Poorly documented variable with 99.5% of its values missing
        -WEEKLNGPCT
        )

# NAs in the dataset
missingvalues <- function(df) {
  NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
  NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

  tibble(
    NAs = NAs, # Not applicable
    NRs = NRs, # Not reported
    perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
  )
}
missingvalues(rhfs)
```

```
## # A tibble: 1 x 3
##      NAs    NRs perc.missing
##    <int>  <int>        <dbl>
## 1 170275 144234        0.402
```

```
# Missing values by interview type
rhfs |>
  summarize(
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )
```

```
##                                           OUTCOME   NAs   NRs perc.missing
## 1                    Interview Compl; field interview 97135 73188    0.3933920
## 2                Sufficient partial; field interview 29948 36570    0.5177304
## 3                     Interview Compl; self response 37769 21028    0.3230888
## 4          Conversion from partial to full interview  3442  7740    0.4538149
## 5                    Military limited field interview   297   839    0.7171717
## 6 Assisted living limited interview; field interview  1286  3761    0.7169034
## 7   Assisted living limited interview; self response   398  1108    0.7130682
```

```
# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to the corresponding REGPAY values.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
      # In a given observation, if any of these columns contain a value other than
```

```
      # -8 or -9 return True.
      if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1


trivially.missing.ownothprop <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OWNOTHPROP == -9 &
      if_any(c(NUMOTHPROP, NUMOTHUNITS), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, NUMOTHPROP, NUMOTHUNITS) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.ownothprop] <- 1


trivially.missing.rntappfeewaiv <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    RNTAPPFEEWAIV == -9 &
      if_any(c(PCTFEEWAIV_R), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, PCTFEEWAIV_R) |>
  pull(id)

rhfs$RNTAPPFEEWAIV[trivially.missing.rntappfeewaiv] <- 1


# Total rental receipts as a percent of property value (TRENTRCTPV_R) is
# calculated with (TOTCOLL + COMMRENT) / MRKTVAL. If COMMSPACE = 2, then
# COMMRENT = 0, so TRENTRCTPV_R = TOTCOLL / MRKTVAL as derived below.
trivially.missing.trentrctpv_r <- rhfs |>
  mutate(id = row_number()) |>
  select(id, TRENTRCTPV_R, COMMSPACE, COMMRENT, RECCOMMREN, TOTCOLL, MRKTVAL_R) |>
  filter(TRENTRCTPV_R == -8 & COMMSPACE == 2 &
      if_all(c(TOTCOLL, MRKTVAL_R), \(x) !x %in% c(-8, -9))
    ) |>
  pull(id)

rhfs$TRENTRCTPV_R[trivially.missing.trentrctpv_r] <-
  rhfs$TOTCOLL[trivially.missing.trentrctpv_r] /
  rhfs$MRKTVAL_R[trivially.missing.trentrctpv_r]


# When CASHDWN is 2, CASHDWNAMT is indeed always -9
rhfs |>
  mutate(id = row_number()) |>
  select(id, CASHDWN, CASHDWNAMT) |>
```

```
  filter(CASHDWN == 2 & CASHDWNAMT != -9)
```

```
## [1] id         CASHDWN    CASHDWNAMT
## <0 rows> (or 0-length row.names)
```

```
# If a CASHDWN payment wasn't made, then the CASHDWNAMT should be 0
rhfs$CASHDWNAMT <- ifelse(rhfs$CASHDWN == 2, 0, rhfs$CASHDWNAMT)


# Same idea as with CASHDWN/CASHDWNAMT
rhfs |>
  mutate(id = row_number()) |>
  select(id, WEEKLNG, WEEKLNGNUM) |>
  filter(WEEKLNG == 2 & WEEKLNGNUM != -9)
```

```
##     id WEEKLNG WEEKLNGNUM
## 1 2173       2          0
## 2 3507       2          0
## 3 4197       2          0
```

```
# If no units are offered for daily/weekly rental, then WEEKLNGNUM (the count
# of those that are offered) should be 0
rhfs$WEEKLNGNUM <- ifelse(rhfs$WEEKLNG == 2, 0, rhfs$WEEKLNGNUM)


# Viewing targets for imputation
rhfs |>
  mutate(id = row_number()) |>
  select(COMMSPACE, COMMRENT, RECCOMMREN) |>
  filter(COMMSPACE == 2)
```

```
##      COMMSPACE COMMRENT RECCOMMREN
## 1            2       -9         -9
## 2            2       -9         -9
## 3            2       -9         -9
## 4            2       -9         -9
## 5            2       -9         -9
## 6            2       -9         -9
## 7            2       -9         -9
## 8            2       -9         -9
## 9            2       -9         -9
## 10           2       -9         -9
## 11           2       -9         -9
## 12           2       -9         -9
## 13           2       -9         -9
## 14           2       -9         -9
## 15           2       -9         -9
## 16           2       -9         -9
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## 18           2       -9         -9
## 19           2       -9         -9
## 20           2       -9         -9
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## 21          2       -9        -9
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## 3458        2       -8        -9
## 3459        2       -8        -9
## 3460        2       -8        -9
## 3461        2       -8        -9
## 3462        2       -8        -9
## 3463        2       -8        -9
## 3464        2       -8        -9
## 3465        2       -8        -9
## 3466        2       -8        -9
## 3467        2       -8        -9
## 3468        2       -8        -9
## 3469        2       -8        -9
## 3470        2       -8        -9
## 3471        2       -8        -9
## 3472        2       -8        -9
## 3473        2       -8        -9
## 3474        2       -8        -9
## 3475        2       -8        -9
## 3476        2       -8        -9
```

```
## 3477        2       -8        -9
## 3478        2       -8        -9
## 3479        2       -8        -9
## 3480        2       -8        -9
## 3481        2       -8        -9
## 3482        2       -8        -9
## 3483        2       -8        -9
## 3484        2       -8        -9
## 3485        2       -8        -9
## 3486        2       -8        -9
## 3487        2       -8        -9
## 3488        2       -8        -9
## 3489        2       -8        -9
## 3490        2       -8        -9
## 3491        2       -8        -9
## 3492        2       -8        -9
## 3493        2       -8        -9
## 3494        2       -8        -9
## 3495        2       -8        -9
## 3496        2       -8        -9
## 3497        2       -8        -9
## 3498        2       -8        -9
## 3499        2       -8        -9
## 3500        2       -8        -9
## 3501        2       -8        -9
## 3502        2       -8        -9
## 3503        2       -8        -9
## 3504        2       -8        -9
## 3505        2       -8        -9
## 3506        2       -8        -9
## 3507        2       -8        -9
## 3508        2       -8        -9
## 3509        2       -8        -9
## 3510        2       -8        -9
## 3511        2       -8        -9
## 3512        2       -8        -9
## 3513        2       -8        -9
## 3514        2       -8        -9
## 3515        2       -8        -9
## 3516        2       -8        -9
## 3517        2       -8        -9
## 3518        2       -8        -9
## 3519        2       -8        -9
## 3520        2       -8        -9
## 3521        2       -8        -9
## 3522        2       -8        -9
## 3523        2       -8        -9
## 3524        2       -8        -9
## 3525        2       -8        -9
## 3526        2       -8        -9
## 3527        2       -8        -9
## 3528        2       -8        -9
## 3529        2       -8        -9
## 3530        2       -8        -9
```

```
## 3531        2       -8        -9
## 3532        2       -8        -9
## 3533        2       -8        -9
## 3534        2       -8        -9
## 3535        2       -8        -9
## 3536        2       -8        -9
## 3537        2       -8        -9
## 3538        2       -8        -9
## 3539        2       -8        -9
## 3540        2       -8        -9
## 3541        2       -8        -8
## 3542        2       -8        -9
## 3543        2       -8        -9
## 3544        2       -8        -9
## 3545        2       -8        -9
## 3546        2       -8        -9
## 3547        2       -8        -9
## 3548        2       -8        -9
## 3549        2       -8        -9
## 3550        2       -8        -9
## 3551        2       -8        -9
## 3552        2       -8        -8
## 3553        2       -8        -9
## 3554        2       -8        -9
## 3555        2       -8        -9
## 3556        2       -8        -9
## 3557        2       -8        -9
## 3558        2       -8        -9
## 3559        2       -8        -9
## 3560        2       -8        -9
## 3561        2       -8        -9
## 3562        2       -8        -9
## 3563        2       -8        -9
## 3564        2       -8        -9
## 3565        2       -8        -9
## 3566        2       -8        -9
## 3567        2       -8        -9
## 3568        2       -8        -9
## 3569        2       -8        -9
## 3570        2       -8        -9
## 3571        2       -8        -9
## 3572        2       -8        -9
## 3573        2       -8        -9
## 3574        2       -8        -9
## 3575        2       -8        -9
## 3576        2       -8        -9
## 3577        2       -8        -9
## 3578        2       -8        -9
## 3579        2       -8        -9
## 3580        2       -8        -9
## 3581        2       -8        -9
## 3582        2       -8        -9
## 3583        2       -8        -9
## 3584        2       -8        -9
```

```
## 3585        2       -8        -9
## 3586        2       -8        -9
## 3587        2       -8        -9
## 3588        2       -8        -9
## 3589        2       -8        -9
## 3590        2       -8        -9
## 3591        2       -8        -9
## 3592        2       -8        -9
## 3593        2       -8        -9
## 3594        2       -8        -9
## 3595        2       -8        -9
## 3596        2       -8        -9
## 3597        2       -8        -9
## 3598        2       -8        -9
## 3599        2       -8        -9
## 3600        2       -8        -9
## 3601        2       -8        -9
## 3602        2       -8        -9
## 3603        2       -8        -9
## 3604        2       -8        -9
## 3605        2       -8        -9
## 3606        2       -8        -9
## 3607        2       -8        -9
## 3608        2       -8        -9
## 3609        2       -8        -9
## 3610        2       -8        -9
## 3611        2       -8        -9
## 3612        2       -8        -9
## 3613        2       -8        -9
## 3614        2       -8        -9
## 3615        2       -8        -9
## 3616        2       -8        -9
## 3617        2       -8        -9
## 3618        2       -8        -9
## 3619        2       -8        -9
## 3620        2       -8        -9
## 3621        2       -8        -9
## 3622        2       -8        -9
## 3623        2       -8        -9
## 3624        2       -8        -9
## 3625        2       -8        -9
## 3626        2       -8        -9
## 3627        2       -8        -9
## 3628        2       -8        -9
## 3629        2       -8        -9
## 3630        2       -8        -9
## 3631        2       -8        -9
## 3632        2       -8        -9
## 3633        2       -8        -8
## 3634        2       -8        -8
## 3635        2       -8        -9
## 3636        2       -8        -9
## 3637        2       -8        -9
## 3638        2       -8        -9
```

```
## 3639        2       -8        -9
## 3640        2       -8        -9
## 3641        2       -8        -9
## 3642        2       -8        -9
## 3643        2       -8        -9
## 3644        2       -8        -9
## 3645        2       -8        -9
## 3646        2       -8        -9
## 3647        2       -8        -9
## 3648        2       -8        -9
## 3649        2       -8        -9
## 3650        2       -8        -9
## 3651        2       -8        -9
## 3652        2       -8        -9
## 3653        2       -8        -9
## 3654        2       -8        -9
## 3655        2       -8        -9
## 3656        2       -8        -9
## 3657        2       -8        -9
## 3658        2       -8        -9
## 3659        2       -8        -9
## 3660        2       -8        -9
## 3661        2       -8        -9
## 3662        2       -8        -9
## 3663        2       -8        -9
## 3664        2       -8        -9
## 3665        2       -8        -9
## 3666        2       -8        -9
## 3667        2       -8        -9
## 3668        2       -8        -9
## 3669        2       -8        -9
## 3670        2       -8        -9
## 3671        2       -8        -9
## 3672        2       -8        -9
## 3673        2       -8        -9
## 3674        2       -8        -9
## 3675        2       -8        -9
## 3676        2       -8        -9
## 3677        2       -8        -9
## 3678        2       -8        -9
## 3679        2       -8        -9
## 3680        2       -8        -9
## 3681        2       -8        -9
## 3682        2       -8        -9
## 3683        2       -8        -9
## 3684        2       -8        -9
## 3685        2       -8        -9
## 3686        2       -8        -9
## 3687        2       -8        -9
## 3688        2       -8        -9
## 3689        2       -8        -9
## 3690        2       -8        -9
## 3691        2       -8        -9
## 3692        2       -8        -9
```

```
## 3693        2       -8        -9
## 3694        2       -8        -9
## 3695        2       -8        -9
## 3696        2       -8        -9
## 3697        2       -8        -9
## 3698        2       -8        -9
## 3699        2       -8        -9
## 3700        2       -8        -9
## 3701        2       -8        -9
## 3702        2       -8        -9
## 3703        2       -8        -9
## 3704        2       -8        -9
## 3705        2       -8        -9
## 3706        2       -8        -9
## 3707        2       -8        -9
## 3708        2       -8        -9
## 3709        2       -8        -9
## 3710        2       -8        -9
## 3711        2       -8        -9
## 3712        2       -8        -9
## 3713        2       -8        -9
## 3714        2       -8        -9
## 3715        2       -8        -9
## 3716        2       -8        -9
## 3717        2       -8        -9
## 3718        2       -8        -9
## 3719        2       -8        -9
## 3720        2       -8        -9
## 3721        2       -8        -8
## 3722        2       -8        -9
## 3723        2       -8        -9
## 3724        2       -8        -9
## 3725        2       -8        -9
## 3726        2       -8        -9
## 3727        2       -8        -9
## 3728        2       -8        -9
## 3729        2       -8        -9
## 3730        2       -8        -9
## 3731        2       -8        -9
## 3732        2       -8        -9
## 3733        2       -8        -9
## 3734        2       -8        -9
## 3735        2       -8        -9
## 3736        2       -8        -9
## 3737        2       -8        -9
## 3738        2       -8        -9
## 3739        2       -8        -9
## 3740        2       -8        -9
## 3741        2       -8        -9
## 3742        2       -8        -9
## 3743        2       -8        -9
## 3744        2       -8        -9
## 3745        2       -8        -9
## 3746        2       -8        -9
```

```
## 3747        2       -8        -9
## 3748        2       -8        -9
## 3749        2       -8        -9
## 3750        2       -8        -9
## 3751        2       -8        -9
## 3752        2       -8        -9
## 3753        2       -8        -9
## 3754        2       -8        -9
## 3755        2       -8        -9
## 3756        2       -8        -9
## 3757        2       -8        -9
## 3758        2       -8        -9
## 3759        2       -8        -9
## 3760        2       -8        -9
## 3761        2       -8        -9
## 3762        2       -8        -9
## 3763        2       -8        -9
## 3764        2       -8        -9
## 3765        2       -8        -9
## 3766        2       -8        -9
## 3767        2       -8        -9
## 3768        2       -8        -9
## 3769        2       -8        -9
## 3770        2       -8        -9
## 3771        2       -8        -9
## 3772        2       -8        -9
## 3773        2       -8        -9
## 3774        2       -8        -9
## 3775        2       -8        -9
## 3776        2       -8        -9
## 3777        2       -8        -9
## 3778        2       -8        -9
## 3779        2       -8        -9
## 3780        2       -8        -9
## 3781        2       -8        -9
## 3782        2       -8        -9
## 3783        2       -8        -9
## 3784        2       -8        -9
## 3785        2       -8        -9
## 3786        2       -8        -8
## 3787        2       -8        -9
## 3788        2       -8        -9
## 3789        2       -8        -9
## 3790        2       -8        -9
## 3791        2       -8        -9
## 3792        2       -8        -9
## 3793        2       -8        -9
## 3794        2       -8        -9
## 3795        2       -8        -9
## 3796        2       -8        -9
## 3797        2       -8        -9
## 3798        2       -8        -8
## 3799        2       -8        -9
## 3800        2       -8        -9
```

```
## 3801         2       -8         -9
## 3802         2       -8         -9
## 3803         2       -8         -9
## 3804         2       -8         -9
## 3805         2       -8         -9
## 3806         2       -8         -9
## 3807         2       -8         -9
## 3808         2       -8         -9
## 3809         2       -8         -9
## 3810         2       -8         -9
## 3811         2       -8         -9
## 3812         2       -8         -9
## 3813         2       -8         -9
## 3814         2       -8         -9
## 3815         2       -8         -9
## 3816         2       -8         -9
## 3817         2       -8         -9
```

```
# If there is no commercial space on this property, then the property did not
# receive any rental income from a commercial use of the property.
rhfs$RECCOMMREN[rhfs$COMMSPACE == 2] <- "2"
rhfs$COMMRENT <- ifelse(rhfs$COMMSPACE == 2, 0, rhfs$COMMRENT)


rhfs |>
  mutate(id = row_number()) |>
  select(RECCOMMREN, COMMRENT) |>
  filter(RECCOMMREN == 2) |>
  summarize(
    n(),
    .by = COMMRENT
  )
```

```
##   COMMRENT  n()
## 1        0 3874
## 2       -9    9
```

```
# If commercial rent was not received, then commercial rent should be 0
rhfs$COMMRENT <- ifelse(rhfs$RECCOMMREN == 2, 0, rhfs$COMMRENT)


# More informative than "-8"
rhfs <- rhfs |>
  mutate(
    UNITSTAT = fct_recode(
      UNITSTAT,
      "Not a single unit" = "-8"
    )
  )


# TOTRENTVAC's (total units for rent or vacant-for rent) 377 missing values are
# due to them being outside its universe of NUMUNITS >= 2. UNITSTAT (status of
```

```r
# single unit) indicates that all of these values are rented or vacant for rent,
# pointing towards the idea that the 377 missing values should have a TOTRENTVAC
# of 1.
table(rhfs$TOTRENTVAC)
```

```
##
##   -9    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
##  377   56  170   88  142   39   84   22   75   19   47   16   54   14   22   26
##   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31
##   46    5   24   17   41   20   14   23   56   15   10   15   29   15   28   19
##   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47
##   30    8   11   25   16   13    8   17   45   12   11    9   17   13   11   11
##   48   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63
##   45   14   33    3    9    5    5    8   20    6    3   11   31    6   10    7
##   64   65   66   67   68   69   70   71   72   73   74   75   76   77   78   79
##   15    6    7    5   14    8   11    6   24    4    5   10    9    3    8    8
##   80   81   82   83   84   85   86   87   88   89   90   91   92   93   94   95
##   25    3    3   12   12    7    7    5   17    7   13    3    4    3   11    6
##   96   97   98   99  100  101  102  103  104  105  106  107  108  109  110  111
##   14    2   12   14   39    7    3    8   10    6    6    4   11    3   10    5
##  112  113  114  115  116  117  118  119  120  121  122  123  124  125  126  127
##   12    7    5    5    6    8    7   11   33    3    6    7    9    3    4    5
##  128  129  130  131  132  133  134  135  136  137  138  139  140  141  142  143
##   12    4   11    5    6    4    5    4   13    5    4    9   13    7    6    5
##  144  145  146  147  148  149  150  151  152  153  154  155  156  157  159  160
##   19    4    5    2   14    7   19    7   10    1    8    5    4    2    4   22
##  161  162  163  164  165  166  167  168  169  170  171  172  173  174  175  176
##    4    4    4    7    1    6    3   12    4    7    9    3    3    4    3   12
##  177  178  179  180  181  182  183  184  185  186  187  188  189  190  191  192
##    4    4    7   18    2    3    6    8    4    6    4   11    3    4    2   11
##  193  194  195  196  197  198  199  200  201  202  203  204  205  206  207  208
##    8    4    2   11    3   12    5   26   10    7    7    6    6    5    5   16
##  209  210  211  212  213  214  215  216  217  218  219  220  221  222  223  224
##    4   11    2    7    3    4    2   18    2    3    6    6    4    5    3    9
##  225  226  227  228  229  230  231  232  233  234  235  236  237  238  239  240
##    2    6    5    7    1    4    2    6    2    4    4    4    3    4    2   27
##  241  242  244  245  246  247  248  249  250  251  252  253  254  255  256  257
##    1    3    6    2    2    4    8    6    8    5   12    1    6    3    6    3
##  258  259  260  261  262  263  264  265  266  267  268  269  270  271  272  273
##    3    5    9    2    6    4    9    2    4    4    9    2    8    1    9    5
##  274  275  276  277  278  279  280  281  282  283  284  285  286  287  288  289
##    2    5    6    3    9    4   10    2    3    3    3    5    3    3   13    1
##  290  291  292  293  294  295  296  297  298  299  300  301  303  304  305  306
##    2    5    7    3    1    2    9    3    2    3   22    6    5   11    1    2
##  307  308  309  310  311  312  313  314  315  316  317  318  319  320  321  322
##    2    5    1    5    7   10    1    2    3    5    4    5    4   12    1    3
##  324  325  326  327  328  329  330  331  332  333  334  335  336  338  339  340
##    8    5    2    1    5    3    1    4    5    2    3    2   11    2    3    5
##  341  344  345  346  347  348  349  350  351  352  353  354  355  356  357  358
##    2    2    4    5    2    7    4    3    6    5    1    2    1    4    4    2
##  359  360  362  363  364  365  366  367  368  369  370  371  372  373  374  375
##    1   12    4    1    4    2    6    2    8    6   11    8   90    1    2    1
##  376  377  378  379  380  382  383  384  385  386  387  388  390  391  392  393
```

```
##    4    2    2    2    1    1    1    3    4    2    1    6    1    1    1    1
##  394  395  396  397  399  400  401  402  404  405  406  408  409  411  412  413
##    2    3    3    1    3    5    1    1    1    2    2    6    1    1    1    1
##  415  416  417  418  419  420  421  422  424  425  426  428  429  430  431  433
##    1    3    1    2    1    1    1    2    1    1    1    2    2    3    1    3
##  434  436  437  440  441  442  444  445  446  447  448  450  452  453  454  456
##    1    2    1    3    1    3    3    1    2    1    3    2    1    1    1    4
##  457  460  461  462  463  464  466  467  469  470  472  476  477  478  480  482
##    1    2    1    1    3    1    1    2    2    1    3    2    2    2    3    1
##  483  484  486  487  488  489  491  492  494  495  496  497  498  500  502  504
##    2    2    2    2    1    1    2    3    1    2    5    1    6    1    3    2
##  506  508  509  510  512  513  514  515  516  517  519  520  521  522  523  524
##    2    2    3    2    1    1    2    1    1    1    1    2    2    1    1    2
##  526  529  531  533  535  540  541  544  545  546  547  550  552  553  557  558
##    2    3    1    1    1    1    2    1    3    2   11    1    1    1    2    2
##  563  565  568  569  574  575  576  579  581  584  587  588  596  597  598  600
##    1    3    1    1    1    1    1    1    1    1    1    3    1    1    1    4
##  604  608  611  614  618  621  622  624  627  630  636  640  642  645  648  653
##    1    1    3    1    1    1    2    1    1    1    1    4    1    1    1    1
##  657  660  664  667  668  669  675  680  684  694  696  698  700  705  706  708
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  709  715  718  720  734  739  741  743  744  752  771  775  779  781  788  789
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  801  802  810  833  839  845  858  862  880  891  900  922 1000 1043 1097 1136
##    2    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## 1137 1215 1236 1241 1274 1293 1296 1322 1324 1350 1354 1416 1444 1474 1530 1730
##    1    1    1    1    2    2    1    1    1    1    1    1    1    1    1    1
## 1817
##    1
```

```
table(rhfs$UNITSTAT)
```

```
##
## Not a single unit                 1                 2
##              4048               360                17
```

```
rhfs$TOTRENTVAC <- ifelse(rhfs$TOTRENTVAC == -9, 1, rhfs$TOTRENTVAC)


# Reverse engineering TOTRENTVAC indicates that it is constructed by NUMUNITS -
# TOWNOC_R - TEMPOC_R - TOTHVA_R (accurate 99.66% of the time). To achieve this
# accuracy, values in TOWNOC_R, TEMPOC_R, and TOTHVA_R are assigned to 0.
rhfs$TOWNOC_R[rhfs$TOWNOC_R %in% c(-9, -8)] <- 0
rhfs$TEMPOC_R[rhfs$TEMPOC_R %in% c(-9, -8)] <- 0
rhfs$TOTHVA_R[rhfs$TOTHVA_R %in% c(-9, -8)] <- 0


# When all components of OPEX_R are explicitly 0, then OPEX_R is 0. The
# corresponding OPEXPU_R (OPEX_R / NUMUNITS_R) are therefore also 0.
trivially.missing.opex_r <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OPEX_R == -9 &
```

```
      if_all(c(OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG, OPPROTH,
               OPSECUR, OPLAND, OPOTHEX), \(x) x == 0)
    ) |>
  select(id, OPEX_R, OPEXPU_R, OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG,
         OPPROTH, OPSECUR, OPLAND, OPOTHEX) |>
  pull(id)

rhfs$OPEX_R[trivially.missing.opex_r] <- 0
rhfs$OPEXPU_R[trivially.missing.opex_r] <- 0


# When TRENOC_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENOC_R, UNITSTAT) |>
  filter(TRENOC_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENOC_R's (number of renter-occupied units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENOC_R <- ifelse(
  # If UNITSTAT is 1, then TRENOC_R is now 1.
  rhfs$UNITSTAT == "1", 1,
  # If UNITSTAT is 2, then TRENOC_R is now 0.
  ifelse(rhfs$UNITSTAT == "2", 0,
         # Otherwise, leave unchanged.
         rhfs$TRENOC_R))


# When TRENVA_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENVA_R, UNITSTAT) |>
  filter(TRENVA_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENVA_R's (number of vacant for rent units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENVA_R <- ifelse(
  # If UNITSTAT is 1, then TRENVA_R is now 0.
```

```
  rhfs$UNITSTAT == "1", 0,
  # If UNITSTAT is 2, then TRENVA_R is now 1.
  ifelse(rhfs$UNITSTAT == "2", 1,
         # Otherwise, leave unchanged.
         rhfs$TRENVA_R))


write.csv(rhfs, "RHFS-2024-Cleaned.csv", row.names = FALSE)
```

```
missing_by_var <- data.frame(
  variable = names(rhfs),
  percent_missing = sapply(
    rhfs,
    \(x) mean(x %in% c(-9))
  )
) |>
  arrange(desc(percent_missing))


# Evidence for MAR (missing at random) in MRKTVAL_R
df |>
  mutate(
    mrktNA = MRKTVAL_R == -9
  ) |>
  summarize(
    survey_mean(mrktNA),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 3
##   NUMCAT_R      coef  `_se`
##   <fct>        <dbl>  <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163
```

```
df |>
  filter(MRKTVAL_R != -9) |>
  summarize(
    mean_mrktval = survey_mean(MRKTVAL_R),
    median_mrktval = survey_median(MRKTVAL_R),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 5
##   NUMCAT_R    mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>              <dbl>           <dbl>          <dbl>             <dbl>
## 1 1 Unit           397050.          42218.         250000            33154.
## 2 2-4 Units        569582.          33304.         425000            50491.
## 3 5-24 Units      2178309.         118540.        2000000           469164.
```

```
## 4 25-49 Units     5000928.         333704.        4000000           224403.
## 5 50+ Units      33895580.        2865340.       20000000          4987831.
```

```r
# For citing R, RStudio, and the packages that were used in this analysis
cite_packages(
  pkgs = "All",
  omit = NULL,
  citation.style = "apa",
  include.RStudio = TRUE,
  cite.tidyverse = FALSE,
  out.format = "docx",
  out.dir = getwd()
)
```

```
##
## Citation report available at C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis

## [1] "C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis-WIP/grateful-report.doc
```